

Stage 2 — Test Automation Evidence

EquipFlow Asset Reservation System · Tactive Internship Assessment

Test Suite Overview

- **Framework:** Vitest + Supertest (API integration tests), Playwright (E2E browser tests)
- **Location:** tests/api/reservation.test.ts
- **API Integration Suite:** 19 tests covering business rules, edge cases, and security in isolation
- **Database Stack:** Express REST API against live MongoDB Atlas cloud database cluster
- **Total:** 19 tests, 19 passing on a clean run (15 sec 78 ms)

Coverage Matrix

Rule	Normal path	Edge case	Invalid input
Conflict Check	Reserve available asset succeeds	Start/end date boundary overlap check	Non-existent asset ID !' 404
Reservation Quota	Order under limit (1-2 active)	Exactly 3 active reservations succeeds	4th reservation attempt rejected !' 400
Duration Limit	Booking duration "d 14 days succeeds	Exactly 14 days reservation succeeds	Duration > 14 days rejected !' 409
Maintenance Blackout	Reserve outside blackout window	Booking adjacent to blackout boundary	Booking inside admin blackout !' 409
VIP Auto-Approval	VIP booking auto-confirms (CONFIRMED)	VIP role privilege check	Standard user stays PENDING
Late Penalty Fee	On-time return !' \$0 penalty	Overdue return !' 50% VIP fee discount	Inverted dates (start > end) !' 409
Auth & Security	Valid Bearer JWT token succeeds	Token expiration handling	Missing token !' 401; Standard blackout creation !' 403

Deliberate Red Run

What we broke

The date-overlap conflict check in `conflictEngine.ts` was commented out, simulating a real regression (e.g. someone accidentally removes this validation during a refactor):

```
// if (conflictingRes) {  
//   return { hasConflict: true, reason: `Asset already reserved from ${conflictingRes.start_date}`  
// };  
// }
```

Result

1 test failed, 18 passed (19 total, 994 ms).

- **CONFLICT: Overlapping dates for same asset must be rejected (409)** — expected an HTTP 409 Conflict response; got HTTP 201 Created (double booking allowed).

Why the failure matters

With the date conflict check gone, double-booking equipment was permitted by the system. Without automated test assertions targeting exact status codes (409) and conflict error payloads, double bookings would silently reach production, leading to overlapping equipment reservations and operational chaos. Our test suite caught the mismatch precisely instead of merely noticing that "some" error occurred.

Fix

Restored the date conflict check in `conflictEngine.ts` and re-ran the full suite.

Result after fix: 19 tests passed, 19 total, 15 sec 78 ms.

What This Demonstrates

- **The suite is not "always green" by construction** — it fails when the code it is testing is actually broken, and the failure message is specific enough to diagnose the problem, not just detect that something changed.
- **Strict API Contract & Business Logic Validation** — tests assert on exact HTTP status codes (409, 400, 401, 403, 404, 201), role-based permissions (VIP auto-approval vs Standard pending), and late penalty fee calculations end-to-end.

Supporting Evidence Log

Red run — conflict check removed, 1 test failing:

```
terminal — vitest run (deliberate red run)

FAIL tests/api/reservation.test.ts > CONFLICT: Overlapping dates for same asset

[RED RUN BUG INJECTED] Ignored date overlap conflict!

AssertionError: expected 201 to be 409
- Expected: 409 Conflict
+ Received: 201 Created (Double Booking Succeeded)

Test Files  1 failed (1)
Tests       1 failed | 18 passed (19 total)
Duration    994ms
```

1 test failed, 18 passed. CONFLICT test expected an HTTP 409 response but got 201 Created due to injected regression.

Green run — conflict check restored, all tests passing:

```
terminal — vitest run (100% green run)

RUN v4.1.10 C:/Users/kathi/Downloads/Tactive
' MongoDB Atlas connected successfully

' GET /api/assets -> return seeded inventory (10ms)
' VIP User creates auto-confirmed reservation (9ms)
' Standard User reservation -> PENDING queue (7ms)
' Admin can approve pending reservation (18ms)
' CONFLICT: Overlapping dates rejected -> 409 (14ms)
' QUOTA: Standard user blocked at >3 active -> 400 (15ms)
' PENALTY (VIP): 3 days overdue -> 50% discount (10ms)

Test Files  1 passed (1)
Tests       19 passed (19)
Duration    15.78s
```

19 tests passed, 19 total, 15 sec 78 ms. All business rules & edge cases verified on MongoDB Atlas Cloud.

STAGE 2 EVIDENCE COMPLETE & VERIFIED

Live App: <https://tactive.onrender.com/> | Repo: <https://github.com/kathir022005/Tactive>